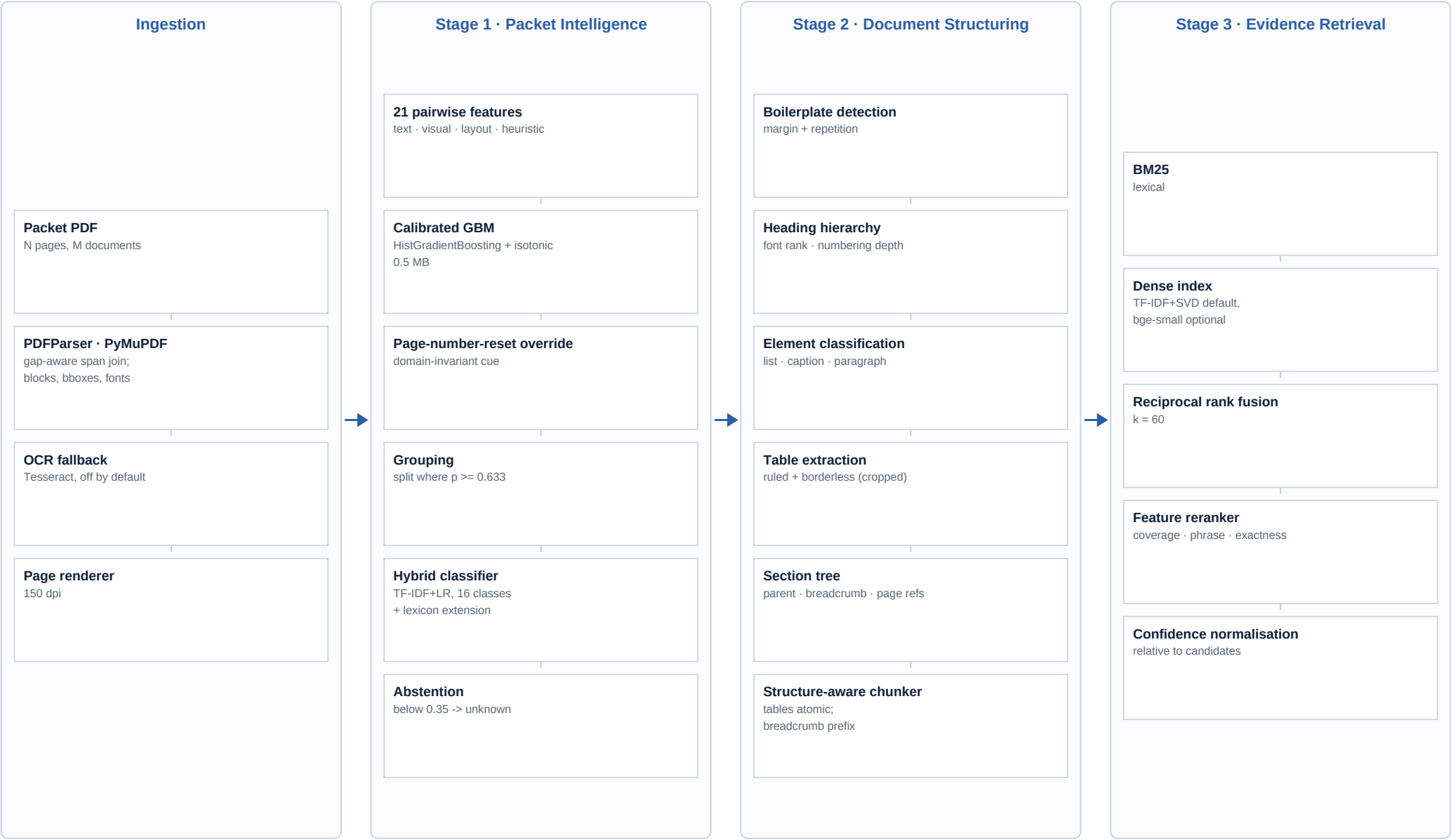


Document Packet Intelligence & Evidence Retrieval

System architecture - three stages with dataclass contracts between them. CPU-only; 5.8 MB of committed models.



Output: evidence + doc_id + page_ref + breadcrumb + confidence | Served by FastAPI: POST /process · POST /retrieve | No GPU, no external API, no network call at inference

Model inventory

Component	Model	Trained on	Size
Boundary classifier	HistGradientBoosting + isotonic calibration	OpenPSS SHORT, 15,906 pairs	0.5 MB
Boundary text signal	TF-IDF vectoriser	same	2.0 MB
Document classifier	TF-IDF + Logistic Regression	RVL-CDIP OCR, 8,887 docs / 16 classes	3.3 MB
Extension classes	Weighted lexicon (no training data exists)	-	< 1 KB
Dense retrieval	TF-IDF + TruncatedSVD	fitted per corpus at index time	112 MB peak RAM
Dense retrieval (optional)	BAAI/bge-small-en-v1.5	pretrained	130 MB

Total committed model footprint: 5.8 MB. No component is trained on the DocSplit benchmark; the optional transformer is a general-purpose pretrained encoder.

Stage contracts

Boundary	Contract
Ingestion -> Stage 1	list[PageRepresentation] - text, blocks, bboxes, fonts, render path
Stage 1 -> Stage 2	list[DocumentGroup] - pages, type, confidence, source, alternatives
Stage 2 -> Stage 3	list[Chunk] - text, page_refs, breadcrumb, element_types, token_count
Stage 3 -> caller	list[EvidenceResult] - evidence, doc_id, page_ref, confidence, scores

Each stage consumes only the contract above it, so a stage can be replaced without touching its neighbours. The Stage 3 dense encoder was swapped three times during benchmarking - hashed bag-of-words, TF-IDF+SVD, then a transformer - with no change to Stage 1 or Stage 2 code.

Artifacts written per packet

File	Contents
pages.json	raw page representations
boundary_features.json	21 features per adjacent page pair
stage1.json	document groups, types, confidences
structured_documents.json	section tree with elements and page refs
markdown/*.md	human-readable rendering per document
chunks.json	retrieval units with breadcrumb and token count
index/	dense vectors and fitted encoder